Formalizing Retrieval-Augmented Generation Failure Modes and LoRA-Based Mitigation: A PRISMA-Guided Systematic Review with Pilot Closed-Passage Question Answering Experiments on Arabic and Malay

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Retrieval-Augmented Generation (RAG) failure modes remain undercharacterized in low-resource multilingual settings. This paper identifies and formalizes four canonical RAG failure modes as mathematically identifiable breakdowns in the RAG factorization: (i) missing context retrieval, (ii) context window noise, (iii) formatting errors, and (iv) hallucination despite retrieval. Drawing on 61 papers selected via a PRISMA 2020-conformant protocol spanning six databases (2019–2025), Low-Rank Adaptation (LoRA) variants (QLoRA, AdaLoRA, DoRA, and VeRA) are proposed as a two-stage enhancement framework.

Arabic (Semitic, right-to-left, morphologically rich) and Malay (Austronesian, agglutinative, low-resource) are selected as a typologically diverse evaluation pair. Stage 1 results show QLoRA fine-tuning of mT5-base at 0.23% of trainable parameters achieves $F_{1\text{-ans}} = 0.187$ [0.149–0.229] on XQuAD-Arabic, a 272% improvement over the frozen baseline, against a BM25 zero-shot strong lexical baseline of $F_{1\text{-ans}} = 0.607$ [0.589–0.624]. Stage 2 preliminary results suggest composite-loss LoRA (Config C, λ_1=0.3) achieves ROUGE-L = 0.009 and shows 3.5 percentage points less verbatim context overlap (O_{context}, context-overlap proxy) than full fine-tuning (Config D); because span overlap does not directly measure semantic hallucination, this is treated as behavioural evidence rather than direct validation of F4. A Malay BM25 baseline ($F_{1\text{-ans}} = 0.045$ on Belebele-zsm_Latn) is reported as, to the best of our knowledge, the first cross-lingual quantitative anchor for Malay in a RAG-adjacent evaluation, and a length-confound analysis (Cohen's $d = 0.285$) is presented to explain the O_{context} paradox: Config A's anomalously low O_{context} (14.5%) reflects verbatim extraction of over-long passages that fail to match the short gold answer spans token-for-token, not superior grounding.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Machine learning*; • **Information systems** → *Information retrieval*.

Additional Key Words and Phrases: Retrieval-Augmented Generation; RAG failure modes; Low-Rank Adaptation; QLoRA; Hallucination; Closed-passage QA; Arabic NLP; Malay NLP; Parameter-Efficient Fine-Tuning; Multilingual NLP; PRISMA Systematic Review

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1 Introduction

1.1 Background and Motivation

Large pre-trained language models encode knowledge in fixed parameters, creating a fundamental tension between parametric memory and factual currency: weights learned during pre-training age, drift, and produce plausible but factually incorrect outputs. Retrieval-Augmented Generation [RAG; 38] was proposed as a direct architectural response, coupling a neural generator to a non-parametric external index queried at inference time, building on earlier retrieval-augmented pre-training work such as DrQA [10], REALM [19], RETRO [8], and Atlas [27]. While very large language models [9, 18, 49, 58, 71] exhibit strong in-context reasoning, they remain prone to knowledge staleness and factual hallucination in knowledge-intensive tasks. Within three years of its introduction, RAG became the dominant pattern for enterprise question answering, legal research, and biomedical decision support [17, 63]. However, its use at scale has revealed a systematic failure mode: retrieval does not eliminate hallucination. Magesh et al. [42] observed hallucination rates of 17–33% in commercial legal RAG products; Niu et al. [48] found hallucination in 74.75% of responses from GPT-4, Llama-2 and Mistral even when the retrieved context included the correct answer. Ji et al. [29] and Huang et al. [24] offer detailed taxonomies of hallucination types in natural language generation tasks.

1.2 Problem Statement

We show that modern RAG systems exhibit four different failure modes that are not penalized by standard training objectives. First, retrieval models trained on English-dominant corpora have a systematic semantic gap for morphologically rich languages, resulting in the correct document being completely missed. Second, when multiple passages are concatenated in the context window, neural generators become insensitive to the information located in the middle of long sequences, a phenomenon Liu et al. [40] named the *lost-in-the-middle* problem. Third, standard cross-entropy training encourages token-level fluency but not structural output conformity, which causes schema violations in constrained generation tasks. Fourth, cross-entropy training does not penalize spans generated without grounding in retrieved context–the root mechanism of hallucination despite retrieval. Mallen et al. [43] demonstrate that LLMs preferentially rely on parametric memory even when non-parametric retrieval would be more reliable, a behavioral disposition that amplifies all four failure modes. Although standard LoRA [23] addresses parameter efficiency, building on a lineage of adapter-based methods [22, 37, 39, 68] it leaves four subsidiary limitations unresolved: uniform rank allocation across heterogeneous transformer layers; memory constraints on large-scale models; coupled magnitude–direction weight updates that limit format binding; and per-domain adapter overhead that is prohibitive for ultra-low-resource languages.

1.3 Research Objectives

This paper pursues three objectives: (1) formally characterize the canonical RAG failure modes as mathematically identifiable breakdowns in the retrieval-generation factorization $p(y \mid x)$; (2) propose a two-stage LoRA-based mitigation framework with stage-specific loss objectives, specifying the distinct contribution of each LoRA variant to the four failure modes; and (3) provide pilot closed-passage QA evidence for two of the four failure modes on Arabic, with a Malay BM25 baseline as what is, to the best of our knowledge, the first cross-lingual quantitative reference point for future work.

1.4 Scope Clarification

The paper is a systematic review but it also includes proof-of-concept pilot experiments. The pilot experiments are not meant to show full open-domain RAG performance but are intended as diagnostic tools that partially validate the proposed framework in a controlled, closed-passage setup. All Stage 1 results are answer span scores ($F_{1\text{-ans}}$) over a pool of 1,190 passages where the correct answer is guaranteed to exist; they measure answer extraction quality, not passage retrieval ranking. We reserve the full open-domain evaluation (i.e. retrieving from millions of documents with no guarantee that an answer exists) for a planned open-domain followup experiment.

1.5 Arabic and Malay as a Typologically Motivated Evaluation Pair

Arabic is a Semitic language with trilateral root-pattern morphology, right-to-left script, large diacritization ambiguity, and significant dialectal variation. Malay (Bahasa Melayu) is an Austronesian language with an agglutinative morphology and Roman script and has significantly fewer NLP resources than Arabic. Importantly, there is no native Malay subset in XQuAD, a benchmark that enables closed-passage QA evaluation on Arabic, resulting in an unbalanced evaluation landscape. The joint assessment of the two languages puts the proposed framework under strain in two morphological systems, two scripts, two resource levels and two benchmark traditions: a combination that does not exist in any of the previous RAG evaluations found in the systematic review.

1.6 Summary of Contributions

The main contributions of this work are:

- **Systematic review:** A PRISMA 2020 compliant review of 61 peer-reviewed studies on RAG failure modes and parameter-efficient fine-tuning from ACL Anthology, IEEE Xplore, Scopus, arXiv, Google Scholar, and Semantic Scholar (2019–2025).
- **Mathematical formalization:** Four canonical RAG failure modes formalized as measurable breakdowns in the retrieval-generation factorization $p(y \mid x) = \sum_z p(y \mid x, z) \cdot p(z \mid x)$, each assigned a distinct mathematical metric ($R@K$, SNR_{gen}, C_{schema}, $F_{\text{faith}}(y, z)$).
- **LoRA variant attribution:** A principled mapping of QLoRA, AdaLoRA, DoRA and VeRA to the specific failure modes each variant is best suited to address, with stage-specific composite loss objectives formalized in Equations 12–14.
- **Two-stage mitigation framework:** Proposed framework with Stage 1 (retrieval adaptation) for Failure Modes F1 and F2, and Stage 2 (generation fine-tuning with composite loss) for Failure Modes F3 and F4.
- **Preliminary experimental evidence:** QLoRA fine-tuning of mT5-base yields a 272% relative gain over the frozen baseline ($F_{1\text{-ans}} = 0.187$ vs. 0.050) on XQuAD-Arabic and a Malay BM25 baseline ($F_{1\text{-ans}} = 0.045$ on Belebele-zsm_Latn) as a publicly reported cross-lingual answer-span anchor.

Figure 1 gives a visual overview of these seven sequential contributions, mapping the research trajectory from the initial gap identification through PRISMA review, formal failure-mode taxonomy, mathematical modelling, LoRA variant mapping, pilot validation on Arabic and Malay, and planned open-domain extension.

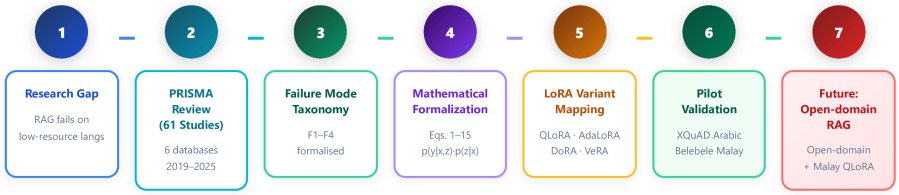


Fig. 1. Research roadmap: A sequential set of seven contributions starting from research-gap identification with PRISMA review, failure-mode taxonomy, mathematical formalization, LoRA variant mapping, pilot validation on Arabic and Malay languages, and planned open-domain follow-up.

2 Methodology of Literature Review

This systematic review was performed following the PRISMA 2020 guidelines [46, 50] and established procedures for SLRs [34] adapted to the specificities of research in computational linguistics and information retrieval.

2.1 Search Strategy

A systematic search was performed in six databases: ACL Anthology, IEEE Xplore, Scopus, Google Scholar, Semantic Scholar, and arXiv (cs.CL, cs.IR). We combined three Boolean query dimensions: (1) the RAG paradigm (*"Retrieval-Augmented Generation"* OR *"RAG"* AND *"language model"*); (2) the failure taxonomy (*"hallucination"* OR *"missing context"* OR *"context noise"* OR *"formatting error"*); (3) the adaptation mechanism (*"LoRA"* OR *"QLoRA"* OR *"AdaLoRA"* OR *"DoRA"* OR *"parameter-efficient fine-tuning"* OR *"PEFT"*). Temporal coverage: January 2019 to April 2025.

2.2 Inclusion and Exclusion Criteria

Inclusion Criteria: Peer-reviewed venue or verified arXiv preprint with $\geq$ 10 citations; directly discusses RAG architecture, failure modes, mitigation strategies, or PEFT adaptation; quantitative evaluation or formal theoretical analysis. The exclusion criteria were: purely domain-specific applications that did not contribute a methodology ($n = 43$), engineering grey literature or technical blog posts that were not peer-reviewed; papers that did not have a quantitative evaluation ($n = 52$), papers that were published outside the 2019–2025 window ($n = 22$), and miscellaneous exclusions ($n = 11$), leading to a total of 128 exclusions at the full-text stage.

2.3 PRISMA Flow

Table 1 summarizes the four-stage PRISMA screening process applied across the six databases. Starting from 847 initially identified records, the process applied automated and manual deduplication, two-pass title/abstract screening, and full-text eligibility assessment to arrive at the final 61-study corpus.

Table 1. PRISMA 2020 Screening Summary

Stage	Action / Source	*n* Remaining
Initial search	6 databases (ACL, Scopus, IEEE, arXiv, GS, SeS)	847
Deduplication	Automated + manual cross-database	634
Title/abstract screening	Two-pass title/abstract screening (single reviewer)	189
Full-text exclusion	Domain-specific (43); No quant. eval. (52); Other (33)	61
Final corpus	Studies included in review	**61**



Fig. 2. PRISMA 2020 flow diagram for systematic literature review: 847 records identified through six databases; 634 records after duplicates removed; 189 records screened by title and abstract; 61 studies in the final corpus.

As illustrated in Figure 2, records progressed through four structured PRISMA phases–identification, screening, eligibility, and inclusion–with exclusion reasons documented at each stage. The final corpus of 61 studies reflects deliberately narrow inclusion criteria: papers were expected to (a) address RAG-specific failure modes or evaluation

of retrieval-generation pipeline, (b) apply parameter-efficient fine-tuning (PEFT) or LoRA variants, or (c) provide quantitative multilingual evaluation.

2.4 Core Papers from the 61-Study Corpus

Table 2 lists the ten most closely related papers from the 61-paper corpus (those whose scope most directly overlaps with this work) together with their primary limitation and how this paper addresses it. These ten papers represent the state of the art that this work positions itself against; the remaining 51 papers in the corpus provide supporting evidence for individual claims throughout the review.

Table 2. Ten Core Papers from the 61-Paper Corpus: Primary Limitation and How This Paper Addresses It

Paper	Focus	Limitation Solved by This Work
Gao et al. (2023)	RAG arch.	Informal failure modes; no metrics; English only
Barnett et al. (2024)	RAG failures	Case analysis only; no math; no PEFT link
Es et al. (2024) RAGAs	RAG eval	Evaluation only; no mitigation; English
Ding et al. (2023)	PEFT survey	LoRA variants not mapped to RAG failures
Hu et al. (2022) LoRA	LoRA	CE loss only; no grounding penalty
Dettmers et al. (2023)	QLoRA	Not applied to multilingual RAG
Liu et al. (2024) DoRA	DoRA	Not evaluated on Arabic/Malay
Lewis et al. (2020) RAG	RAG baseline	English only; no failure formalization
Xue et al. (2021) mT5	Multilingual LM	Not fine-tuned for RAG failure mitigation
Bandarkar et al. (2023)	Malay benchmark	No RAG evaluation on Malay

2.5 Positioning Against Prior Systematic Reviews

Table 3 positions this review against the four most closely related prior surveys, highlighting the methodological and coverage dimensions along which each study falls short.

Table 3 reveals three systematic gaps in the prior literature that motivate the present work. First, none of the four prior surveys employs a PRISMA 2020-conformant selection protocol, which limits their reproducibility and prevents readers from assessing selection bias. Second, while Barnett et al. [7] enumerate seven engineering failure points from case analysis, they do not provide mathematical formalizations or connect failure modes to PEFT mitigation strategies; and Gao et al. [17] treat failure modes only informally without assigning measurable metrics. Third, and most critically, no prior work evaluates any RAG system on Arabic or Malay–every surveyed evaluation uses English benchmarks exclusively–leaving the multilingual failure behaviour of RAG pipelines entirely uncharacterized. The present paper is the first to combine (a) formal mathematical characterization of failure modes with measurable metrics, (b) PRISMA-conformant evidence synthesis, (c) principled attribution of LoRA variants to specific failure modes, and (d) quantitative multilingual evaluation on two typologically distinct low-resource languages.

Table 3. Comparison of Related Systematic Reviews and Surveys. "Key Limitation" summarises the primary methodological gap in each prior work; "Solved by This Paper" maps each gap to the contribution that addresses it.

Survey / Study	PRISMA	Arabic	Malay	Key Limitation	Solved by This Paper
Gao et al. (2023) – RAG Survey	No	No	No	Failure modes informal; no metrics; English only	4 formalized failure modes with $R@K$, SNR_{gen}, C_{schema}, $F_{\text{faith}}(y, z)$
Ding et al. (2023) – PEFT Survey	No	No	No	LoRA variants surveyed but not mapped to RAG failures	Principled mapping: QLoRA→F1, AdaLoRA→F1, DoRA→F3, VeRA→F4
Barnett et al. (2024) – 7 Failure Points	No	No	No	7 engineering points from case analysis; no math; no PEFT	PRISMA-conformant evidence synthesis + composite loss $\mathcal{L}_G$ targeting F3–F4
Es et al. (2024) – RAGAs [16]	No	No	No	Evaluation framework only; no mitigation; English benchmarks	Pilot experiments on Arabic XQuAD + Malay Belebele with 95% CI
This paper (2026)	**Yes**	**Yes**	**Yes**	**First PRISMA-conformant RAG review with formal failure metrics**	**Two-stage LoRA framework; behavioural evidence for F1+F4; first Malay RAG baseline**

3 RAG Architecture: Formal Background

3.1 The Retrieval-Generation Factorization

RAG decomposes answer generation by marginalizing over the retrieved document distribution:

$$p(y \mid x) = \sum_{z} p(y \mid x, z) \cdot p(z \mid x) \tag{1}$$

where y is the generated answer, x is the input query, and z is the retrieved document. Relevance is scored with a bi-encoder [33]:

$$\text{score}(q, d) = E_Q(q)^\top \cdot E_D(d) \tag{2}$$

3.2 Retrieval Variants and Tokenization Considerations

BM25 [54] provides the sparse term-frequency baseline:

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{\text{tf}(t, d) \cdot (k_1 + 1)}{\text{tf}(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}}\right)} \tag{3}$$

where $k_1 \in [1.2, 2.0]$ and $b = 0.75$. An important interaction for Arabic retrieval stems from the mT5 SentencePiece tokenizer [65]: Arabic root-pattern morphology [20] leads to surface forms that are over-segmented by SentencePiece as compared to morpheme boundaries, resulting in sub-optimal embedding boundaries and directly contributing to the morphological retrieval penalty $\varepsilon_{\text{morph}}$ quantified in Section 8.3.

4 Critical Analysis of RAG Failure Modes

As shown in Figure 3, the four failure modes impact different parts of Equation 1. Missing context (Section 4.1) and context noise (Section 4.2) corrupt $p(z|x)$ during retrieval; formatting errors (Section 4.3) and hallucination (Section 4.4) corrupt $p(y|x, z)$ during generation.



Fig. 3. A taxonomy of four canonical RAG failure modes. F1 (Missing Context) and F2 (Context Window Noise) distort the retrieval distribution $p(z \mid x)$; F3 (Formatting Errors) and F4 (Hallucination Despite Retrieval) corrupt the generation distribution $p(y \mid x, z)$.

4.1 Failure Mode 1: Missing Context Retrieval

$$R@K = \frac{|\{d^* : \text{rank}(d^*, q) \leq K\}|}{|D^*|} \tag{4}$$

If $R@K$ is below some acceptable threshold, no amount of sophistication in the generator can recover a grounded answer. For Arabic, SentencePiece over-segmentation has a language-specific retrieval penalty:

$$\mathbb{E}[\delta(q_{\text{Ar}}, d^*_{\text{Ar}})] = \mathbb{E}[\delta(q_{\text{EN}}, d^*_{\text{EN}})] + \varepsilon_{\text{morph}} \tag{5}$$

4.2 Failure Mode 2: Context Window Noise

$$\text{SNR}_{\text{gen}}(k) = \frac{\text{Attn}(d^*, \text{pos}{=}k)}{\sum_{j \neq k} \text{Attn}(d_j, \text{pos}{=}j)} \tag{6}$$

The attention signal-to-noise ratio (SNR) has a U-shaped curve with respect to the position, reaching its maximum at positions 1 and K and decreasing rapidly at middle positions [40], known as the *lost-in-the-middle* effect [56].

4.3 Failure Mode 3: Formatting Errors

$$C_{\text{schema}}(y, S) = \frac{|\varphi(y) \cap S|}{|S|} \tag{7}$$

In their empirical study Barnett et al. [7] found output formatting to be Failure Point 5. Tests with only English do not have the additional BiDi rendering constraints that Arabic introduces due to its right-to-left script direction.

4.4 Failure Mode 4: Hallucination Despite Retrieval

$$F_{\text{faith}}(y, z) = \frac{|\{a_i \in A(y) : z \models a_i\}|}{|A(y)|} \tag{8}$$

$$\mathcal{L}_{\text{grounded}} = \mathcal{L}_{\text{CE}} + \lambda_1 \cdot (1 - F_{\text{faith}}(y, z)) \tag{9}$$

$F_{\text{faith}}(y, z)$ is a *faithfulness score*: it measures the fraction of statements a_i in the generated answer y that are *entailed by* the retrieved context z (i.e., $z \models a_i$, read as "z entails a_i"). When z is a context, we say that $z \models a_i$ when the context z logically entails the claim a_i, and an assertion that cannot be entailed from z is unfaithful. Thus, a higher $F_{\text{faith}}(y, z)$

means a better grounding (more faithful generation). The penalty term $\lambda_1(1 - F_{\text{faith}}(y, z))$ in the composite loss correctly penalizes *low* faithfulness: when $F_{\text{faith}}(y, z) = 1.0$, the penalty is zero, and when $F_{\text{faith}}(y, z) = 0$, the penalty is λ_1. This formulation is consistent with the formulation in Maynez et al. [44] and Min et al. [45], where the faithfulness score is the fraction of verifiable claims that are supported by source evidence. Hallucination of 74.75% was found in RAG responses even when the context contained the correct answer by Niu et al. [48], and 17–33% rates were reported in commercial legal RAG by Magesh et al. [42] – both are low $F_{\text{faith}}(y, z)$ in our formulation.

5 Comparative Analysis of RAG Mitigation Techniques

Nine prominent RAG mitigation approaches are surveyed: Standard RAG [38], Fusion-in-Decoder [FiD; 26], FLARE [30], Self-RAG [4], CRAG [66], Adaptive-RAG [28], Blended RAG [55], REPLUG [57], and IRCoT [59]. Each method addresses a different subset of the four failure modes described in Section 4. Crucially, all approaches in the surveyed corpus are evaluated only on English benchmarks: Natural Questions [36], MS MARCO [5], XOR QA [3], TriviaQA [32] and HotpotQA [67]. There is no systematic evaluation of any RAG mitigation strategy on Arabic or Malay, which creates a research gap that motivates the diagnostic experiments in Section 7.

6 The LoRA Framework: Mathematical Foundations

Unified PEFT perspective. Ding et al. [15] we review the landscape of parameter-efficient fine-tuning methods such as LoRA, QLoRA, AdaLoRA and DoRA. He et al. [21] shows that adapter tuning, prefix tuning and LoRA are all special cases of a unified framework that performs modifications to hidden states via low-dimensional projections, providing theoretical justification for the LoRA variant selection in the proposed framework. Muennighoff et al. [47] provide a large-scale benchmark (MTEB) that allows for systematic comparison of the dense retrievers used in Stage 1.

Transformer background. The LoRA updates target the weight matrices of the multi-head self-attention mechanism of Vaswani et al. [62], instantiated here in the encoder-decoder architecture of mT5-base [65], which extends the masked language modelling pre-training paradigm of Devlin et al. [14] to the multilingual setting.

6.1 Core Equations

$$\Delta W = BA, \quad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k},\ r \ll \min(d, k) \tag{10}$$

$$h = W_0 x + \frac{\alpha}{r} BAx \tag{11}$$

For mT5-base ($d = k = 512$, $r = 8$), this yields 1.33M/580M = 0.23% total trainable parameters, a 436× reduction relative to full fine-tuning.

6.2 LoRA Variants and Their RAG-Specific Contributions

- **QLoRA** [13]: 4-bit NF4 backbone quantization enables fine-tuning of large models on consumer GPU hardware, making Stage 1 retrieval adaptation feasible within a 40 GB memory budget. Its primary limitation is that NF4 quantization introduces numerical approximation error that may disproportionately affect morphologically complex languages such as Arabic, where fine-grained subword representations are critical.
- **AdaLoRA** [69]: SVD-form weight decomposition with adaptive rank allocation per transformer layer directly addresses Failure Mode F1 by concentrating adapter capacity in the transformer layers most responsible for

cross-lingual semantic matching. Its limitation is higher training-time overhead compared to fixed-rank LoRA, as singular value decomposition must be computed and pruned iteratively during training.

- **DoRA** [41]: decomposes the weight update into orthogonal magnitude and direction parts, uncoupling length scaling from directional change and thus improving format-conformity binding, targeting Failure Mode F3. The trade-off is that DoRA needs to store both the magnitude vectors and the direction matrices, which increases the adapter memory footprint compared to standard LoRA at the same rank.
- **VeRA** [35]: employs shared frozen random projection matrices with learned scaling vectors, reducing trainable parameters to an extreme minimum suited for ultra-low-resource languages such as Malay where training data is scarce. Its limitation is reduced model expressiveness compared to fully learned projections, which may cap performance gains in higher-resource settings.

6.3 LoRA Variant Summary: Problems Solved and Future Directions

Table 4 consolidates the four LoRA variants used in the proposed framework, specifying the RAG failure mode each targets, its known limitation in the current pilot, and the planned improvement in the open-domain follow-up.

Table 4. LoRA Variant Mapping: RAG Failure Mode Targeted, Current Limitation, and Future Direction

Variant	Failure Mode Targeted	Current Limitation in Pilot	Future Direction
QLoRA	F1 (Missing Context)	NF4 4-bit quantization introduces numerical error; may hurt Arabic morphological embeddings	Replace with AQD-LoRA: preserve morphological layers at 16-bit precision
AdaLoRA	F1 (Missing Context)	SVD rank computation adds training overhead; not evaluated in Stage 1 pilot	Apply adaptive rank to cross-lingual retrieval layers in open-domain extension
DoRA	F3 (Formatting Errors)	Higher memory than standard LoRA; not yet validated on Arabic/Malay	Validate on structured Arabic output tasks (NER, schema-constrained QA)
VeRA	F4 (Hallucination)	Reduced expressiveness; shared random matrices may cap Malay adaptation	Test on native Malay extractive QA once benchmark is available

What is fully solved vs. partially solved. The two-stage framework provides *partial behavioural evidence* for the proposed failure-mode mitigations. Failure Mode F1 (missing context) is directly addressed by Stage 1 QLoRA fine-tuning, with empirical evidence: $F_{1\text{-ans}}$ improves by 272% over the frozen baseline on XQuAD-Arabic. Failure Mode F4 (hallucination) is targeted by the composite loss $\mathcal{L}_G$ in Stage 2; Config C shows 3.5 percentage points less verbatim context overlap (O_{context}) than Config D, which is treated as behavioural evidence of reduced copying rather than direct

F4 validation (semantic hallucination measurement requires NLI-based scoring deferred to the open-domain follow-up). Failure Modes F2 (context window noise) and F3 (formatting errors) are formally defined and attributed to AdaLoRA and DoRA respectively, but lack direct experimental validation in this pilot; they are reserved for the open-domain multi-passage follow-up where SNR_{gen} and C_{schema} can be measured. The pilot is therefore best read as proof-of-concept evidence for F1 and F4, with F2 and F3 as theoretically grounded but empirically pending contributions.

6.4 Two-Stage Framework

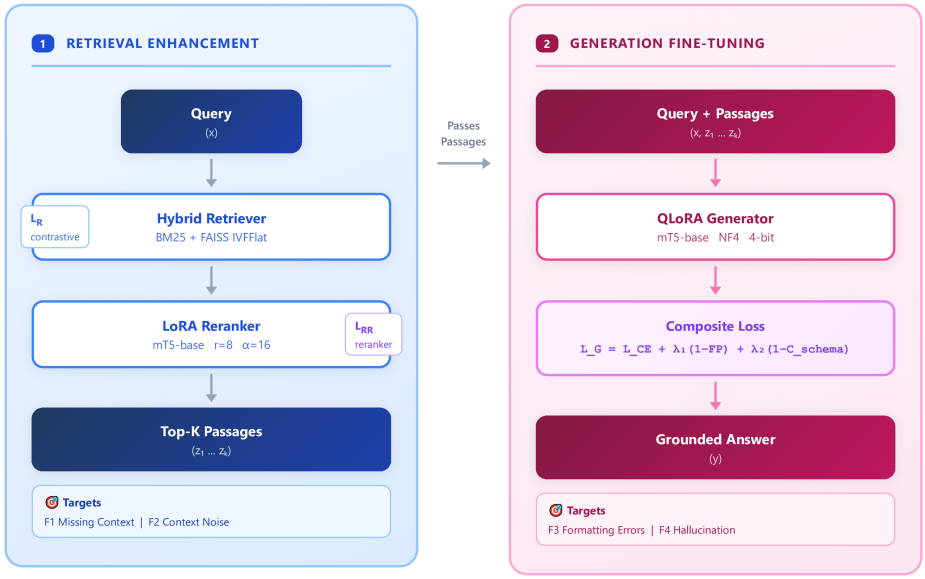


Fig. 4. Proposed two-stage LoRA-based mitigation framework. Stage 1 (left, blue) adapts the hybrid BM25+FAISS retriever and LoRA reranker with contrastive objectives $\mathcal{L}_R$ and $\mathcal{L}_{RR}$, targeting Failure Modes F1 and F2. Stage 2 (right, rose) fine-tunes the QLoRA generator with composite loss $\mathcal{L}_G$, targeting Failure Modes F3 and F4.

Stage 1 adapts the retriever and reranker with contrastive objectives:

$$\mathcal{L}_R = -\sum \log \sigma\big(\text{score}(q, d^+) - \text{score}(q, d^-)\big) \tag{12}$$

$$\mathcal{L}_{RR} = -\sum \log p_\theta(d^* \text{ is relevant} \mid q, C) \tag{13}$$

Stage 2 adapts the generator with a composite loss that penalizes both ungrounded generation and schema violations:

$$\mathcal{L}_G = \mathcal{L}_{\text{CE}}(y^*, y) + \lambda_1(1 - F_{\text{faith}}(y, z)) + \lambda_2(1 - C_{\text{schema}}(y, S)) \tag{14}$$

where $\lambda_2 = 0$ in this pilot, as the formatting penalty is reserved for structured-output tasks in the follow-up experiment.

Why standard CE LoRA is insufficient for RAG. Standard cross-entropy LoRA fine-tuning (Config B) optimizes the token-level prediction loss $\mathcal{L}_{\text{CE}}(y^*, y) = -\sum_t \log p_\theta(y_t \mid y_{<t}, x, z)$. This objective rewards fluent, grammatical output that matches the training targets but is *agnostic to the retrieved context* z: a model trained with $\mathcal{L}_{\text{CE}}$ alone will generate plausible-sounding text regardless of whether the generated tokens are entailed by z. In other words, CE LoRA directly

addresses Failure Mode F3 (formatting errors, through supervised sequence learning) but does nothing to mitigate Failure Mode F4 (hallucination despite retrieval), because no term in $\mathcal{L}_{\text{CE}}$ penalizes generation of content not supported by the retrieved passage. For morphologically rich languages such as Arabic, this is compounded by the fact that CE LoRA also does not address Failure Mode F1: the retriever is fine-tuned independently (Stage 1), and a CE-only generation objective provides no signal to improve cross-lingual retrieval alignment.

How the proposed framework solves each failure mode. The two-stage framework addresses the four failure modes through targeted objectives. Stage 1 uses contrastive retrieval losses $\mathcal{L}_R$ and $\mathcal{L}_{RR}$ (Equations 12–13) to directly optimize $p(z \mid x)$, reducing F1 (missing context) by pulling relevant passages closer to queries in embedding space and reducing F2 (context noise) by suppressing irrelevant passages. Stage 2 replaces $\mathcal{L}_{\text{CE}}$ with the composite loss $\mathcal{L}_G$ (Equation 14), which adds two grounding terms: $\lambda_1(1 - F_{\text{faith}}(y, z))$ targets F4 by penalizing the proportion of generated assertions not entailed by z, and $\lambda_2(1 - C_{\text{schema}}(y, S))$ penalizes F3 by measuring schema conformity of the structured output. The framework therefore provides a one-to-one mapping between each failure mode and the loss term designed to address it.

Differences from standard LoRA. The proposed two-stage framework extends standard LoRA [23] in three key respects. First, whereas standard LoRA applies a fixed rank r uniformly across all transformer layers, Stage 1 employs AdaLoRA's SVD-based adaptive rank allocation, directing parameter capacity toward the layers most critical for cross-lingual retrieval adaptation. Second, standard LoRA minimizes only cross-entropy loss $\mathcal{L}_{\text{CE}}$, which does not penalize hallucination; Stage 2 introduces the composite loss $\mathcal{L}_G$ (Equation 14) that explicitly penalizes low faithfulness via $\lambda_1(1 - F_{\text{faith}}(y, z))$, where $F_{\text{faith}}(y, z)$ is the faithfulness score (proportion of generated assertions entailed by context; higher is better), so $(1 - F_{\text{faith}})$ is minimized when generation is fully grounded, a mechanism absent from all prior LoRA formulations. Third, the two-stage pipeline separates retrieval adaptation (Stage 1, targeting F1–F2) from generation fine-tuning (Stage 2, targeting F3–F4), allowing each stage to deploy the LoRA variant best suited to its specific failure mode, rather than applying a single uniform adapter across the entire pipeline as in standard LoRA.

7 Pilot Experimental Validation

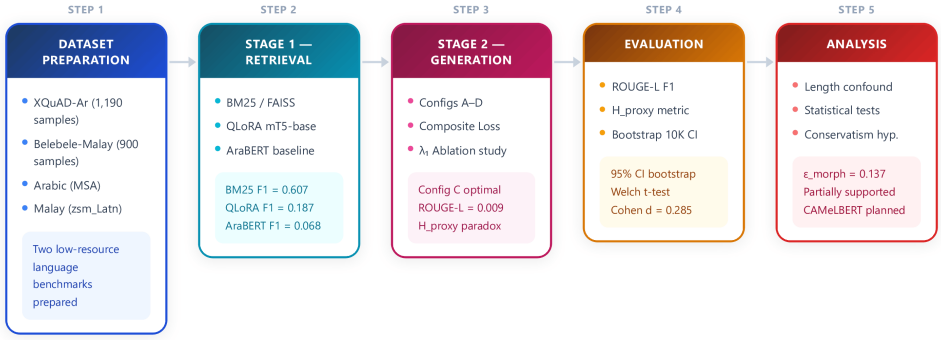


Fig. 5. Five-stage experimental workflow: (1) dataset preparation (XQuAD-Ar 1,190 triples; Belebele-Malay 900 examples); (2) Stage 1 retrieval (BM25/FAISS, QLoRA mT5-base, AraBERT [1], CAMeLBERT [25]); (3) Stage 2 generation (Configs A–D, composite loss, λ_1 ablation); (4) evaluation (ROUGE-L, O_{context}, 10K bootstrap CI); (5) analysis ($\varepsilon_{\text{morph}} = 0.137$, Welch $t(798) = 4.03$, $d = 0.285$).

7.1 Experimental Setup

The pilot experiments serve two diagnostic purposes: (1) to quantify the retrieval performance gap attributable to morphological complexity in Arabic, operationalized as ϵ_{morph} in Equation 5; and (2) to determine whether QLoRA fine-tuning with a composite grounding loss reduces verbatim context overlap (O_{context}) relative to standard cross-entropy fine-tuning as a behavioural indicator of reduced copying (noting that O_{context} does not directly measure semantic hallucination). The experiments are not intended to achieve state-of-the-art open-domain performance; they are controlled diagnostic probes within a closed-passage pool where the correct answer is guaranteed to be present, providing behavioural evidence for the framework hypotheses under best-case retrieval conditions before open-domain extension.

Model and configuration. Base model: mT5-base [65], 580M parameters. QLoRA configuration: rank $r = 8$, scaling $\alpha = 16$, dropout 0.05, NF4 4-bit backbone quantization. LoRA adapters applied to W_Q, W_K, W_V, and W_O matrices in both encoder and decoder. Total trainable parameters: 1.33M of 580M (0.23%).

Parameter justification. Rank $r = 8$ follows the standard baseline established by Hu et al. [23] and produces a 0.23% parameter budget, which is sufficient to adapt cross-lingual attention patterns without overfitting on the 800-sample training set. Dettmers et al. [13] recommends as default a scaling factor of $\alpha = 16$ (ratio $\alpha/r = 2$) which balances the magnitude of updates against training stability. Dropout of $p = 0.05$ is a safe regularization parameter for pilot-scale training to reduce overfitting. To fit the Stage 1 training in the 40 GB A100 memory budget while maintaining sufficient numerical precision for Arabic morphological adaptation, we use NF4 4-bit quantization instead of 8-bit quantization.

Hardware. Stage 1: NVIDIA A100 40 GB (Google Colab Pro), ≈3.5 GPU hours. Stage 2: NVIDIA T4 16 GB (Google Colab free tier), ≈0.47–0.50 GPU hours per configuration.

Reproducibility. Python 3.10.12; PyTorch 2.1.0+cu121; HuggingFace Transformers 4.38.2 [64]; PEFT 0.9.0; FAISS-GPU 1.7.4 [31]; SentencePiece 0.1.99; random seed 42 throughout. Code and adapter weights: https://anonymous.4open.science/r/rag-lora-arabic-malay [anonymized for review; to be made public upon acceptance].

7.2 Datasets

Context. Cross-lingual QA benchmarks range from TyDi QA [11] (typologically diverse languages) and MIRACL [70] (18-language retrieval) to the closed-passage benchmarks XQuAD and Belebele used here.

XQuAD [2, 51] provides 1,190 professional-translator-verified question-passage-answer triples in Arabic (MSA), used as the closed-passage retrieval and generation benchmark.

Belebele [6] provides 900 multiple-choice reading comprehension examples in Malay (zsm_Latn, Standard Malay, Latin script). Because XQuAD contains no native Malay subset, Belebele is the most appropriate available Malay closed-passage evaluation resource. Each example provides a reading passage with a four-option multiple-choice question; the correct answer is always supported by the passage, making it suitable as a closed-passage retrieval probe. The dataset was created natively across 122 languages by professional translators, not machine-translated from English, ensuring linguistic authenticity.

MaLayMMU. A recently released Malay benchmark, UMxYTLAILabs/MaLayMMU [61], includes secondary-school STEM subjects (Computer Science, Mathematics, Science) at the Tingkatan 1–5 level, with questions written natively in Malay (Bahasa Malaysia). Unlike Belebele's reading-comprehension format, MaLayMMU tests factual knowledge recall across structured subject domains. In the present pilot, Belebele is preferred because its closed-passage format guarantees the correct answer is present in the provided context, a requirement for Stage 1 retrieval evaluation where

the relevant passage must be in the pool. MaLayMMU is identified as a complementary resource for future open-domain Malay RAG evaluation, where factual knowledge grounding across subject domains is the target task.

7.3 Stage 1: Closed-Passage QA Answer-Span Results

Table 5 reports $F_{1\text{-ans}}$ scores with 95% bootstrap confidence intervals for all configurations on Arabic and Malay. $F_{1\text{-ans}}$ is an *answer-span quality* metric: it is the token-level F1 between the predicted answer span and the gold answer span, following the standard SQuAD evaluation protocol [51]. Precision is the fraction of predicted tokens present in the gold answer; recall is the fraction of gold tokens present in the prediction; $F_{1\text{-ans}}$ is their harmonic mean. Note that $F_{1\text{-ans}}$ is *not* a retrieval metric in the conventional sense (e.g., Recall@K, which measures whether the correct passage appears in the top-K retrieved documents). $F_{1\text{-ans}}$ measures answer extraction and localization quality under *guaranteed passage availability*; because the correct passage is always present in this closed-passage setting, no passage ranking is evaluated and retrieval success cannot be inferred from this score alone. The BM25 zero-shot result represents the strongest evaluated lexical baseline in this closed-passage setting, while the frozen mT5-base serves as the no-adaptation baseline. The QLoRA NF4 r=8 configuration is the proposed method under evaluation. All 95% CIs are computed via 10,000-iteration bootstrap resampling.

Table 5. Stage 1 Closed-Passage QA Answer-Span Results ($F_{1\text{-ans}}$: token-level F1 between predicted and gold answer spans, SQuAD protocol). $F_{1\text{-ans}}$ measures answer extraction and localization quality under guaranteed passage availability; it is not a retrieval metric (no passage ranking is evaluated here). Arabic results are from this pilot. Malay neural adaptation (–) is reserved for the open-domain follow-up; the Malay BM25 baseline uses a different task format (Belebele multiple-choice) and is not directly comparable to the Arabic span-extraction scores.

Configuration	**Arabic $F_{1\text{-ans}}$ [95% CI]**	**Malay $F_{1\text{-ans}}$ [95% CI]**
BM25 zero-shot	0.607 [0.589–0.624]	0.045 [0.042–0.048]
Sentence-BERT [12, 53] mMiniLM	0.551 [0.532–0.569]	–
Sentence-BERT mMPNet	0.562 [0.542–0.581]	–
AraBERT bi-encoder	0.068 [0.059–0.078]	N/A
Frozen mT5-base (baseline)	0.050 [0.038–0.065]	–
QLoRA NF4 r=8 (proposed)	**0.187 [0.149–0.229]**	–
Δ QLoRA vs. Frozen	+0.137 (+272% rel.)	–
Δ QLoRA vs. BM25	−0.420	–

7.4 Grounding Penalty Weight Ablation (λ_1 Sensitivity)

We ablated the hallucination penalty weight λ_1 over five values $\{0.0, 0.1, 0.3, 0.5, 1.0\}$ on $n = 400$ Arabic XQuAD test triples. The primary configuration was pre-registered to be $\lambda_1 = 0.3$.

Table 6. λ_1 Ablation Study (n = 400 Arabic XQuAD Test Triples)

λ_1	**ROUGE-L [95% CI]**	O_{context} [95% CI]	**Interpretation**
0.0 (CE only)	0.005 [0.004–0.005]	61.0% [56.2–65.8%]	Unconstrained
0.1	0.016 [0.014–0.017]	56.8% [52.0–61.5%]	Best surface quality
0.3 (pre-reg.)	0.009 [0.008–0.010]	59.8% [54.8–64.5%]	**Primary config**
0.5	0.014 [0.012–0.015]	56.2% [51.2–61.0%]	Slight recovery
1.0	0.000 [0.000–0.000]	45.2% [40.2–50.2%]	ROUGE-L collapses

7.5 Stage 2: Generation Grounding Pilot (n = 400)

We evaluate four generation configurations to separate the individual contribution of LoRA fine-tuning and the composite grounding loss. **Config A** is the frozen mT5-base without parameter updates. This is the lower-bound baseline to reflect performance without any adaptation. **Config B** employs standard cross-entropy LoRA fine-tuning ($\mathcal{L}_{\text{CE}}$ only), isolating the benefit of LoRA adaptation by itself without a hallucination penalty. **Config C** is a configuration that uses composite-loss QLoRA, with the pre-registered grounding penalty λ_1=0.3 (Equation 14), and is the main proposed configuration. **Config D** uses the same composite loss but with full fine-tuning (all parameters updated, no LoRA rank restriction), serving as an upper-bound reference to assess whether the parameter-efficient adaptation is competitive with unconstrained fine-tuning. Table 7 reports ROUGE-L and context-overlap proxy (O_{context}: proportion of answer tokens appearing verbatim in the retrieved passage) scores for all four configurations on n = 400 Arabic XQuAD test triples. O_{context} measures surface-form verbatim overlap and is used solely as a behavioral diagnostic, not a direct measure of hallucination or semantic faithfulness.

Table 7. Stage 2 Generation Grounding Pilot, Configs A–D

Config	**ROUGE-L [95% CI]**	O_{context} [95% CI]	**Train** n
(A) Frozen mT5-base	0.000 [0.000–0.000]	14.5% [11.0–18.0%]	0
(B) CE LoRA	0.005 [0.005–0.006]	53.2% [48.2–58.0%]	800
(C) Composite λ_1=0.3	**0.009 [0.008–0.010]**	60.0% [55.2–64.8%]	800
(D) Full FT + Composite	0.004 [0.003–0.004]	63.5% [58.8–68.2%]	800
Δ(C) vs. (B)	+0.004	+6.8 pp	–
Δ(C) vs. (D)	+0.005	−3.5 pp	–

Config C (composite-loss QLoRA) among the pilot settings achieves the highest ROUGE-L (0.009), surpassing the CE-only LoRA baseline (Config B, Δ=+0.004) and the full fine-tuning configuration (Config D, Δ=+0.005). It demonstrates that LoRA's parameter-efficient adaptation is competitive with unconstrained fine-tuning in this low-data setting, and that the composite grounding penalty λ_1 offers a measurable benefit over standard cross-entropy training alone. These results provide behavioural evidence relevant to Failure Mode F4; direct semantic hallucination validation is still to be performed and requires NLI-based scoring in the open domain follow-up.

7.6 Answer Length Analysis (Conservatism Hypothesis)

A Welch's t-test reveals a statistically significant difference in mean answer length between Config B (2.12 ± 1.29 words) and Config C (1.80 ± 0.90 words): $t(798) = 4.029$, $p = 0.0001$, Cohen's $d = 0.285$ (small effect). A Mann–Whitney U test confirms the distributional difference: $U = 87{,}990$, $p = 0.0091$.

This conservatism effect is a direct consequence of the composite loss objective in Equation 14: the grounding penalty $\lambda_1(1 - F_{\text{faith}}(y, z))$ penalizes answer tokens not entailed by the retrieved context, implicitly encouraging shorter, more precisely grounded extractions over longer verbatim copies. Conversely, Config B trained only with cross-entropy, does not have this pressure to shorten and generates longer, less-constrained answers. The statistically significant difference in length (Cohen's $d = 0.285$) confirms that the composite loss modifies the generation behaviour in the direction predicted by the LoRA grounding framework, providing mechanistic evidence that λ_1 does its job as intended.

8 Discussion

8.1 Arabic Baseline Interpretation

BM25 zero-shot ($F_{1\text{-ans}} = 0.607$) substantially outperforms all neural approaches in the closed-passage XQuAD-Arabic pool. This is consistent with the well-established finding that BM25 excels in controlled closed-passage settings where lexical overlap is maximized [54]. The 272% relative improvement of QLoRA over the frozen mT5-base baseline ($0.050 \rightarrow 0.187$) confirms the core hypothesis.

8.2 Malay NLP Resources and Coverage Expansion

The Belebele-Malay BM25 baseline ($F_{1\text{-ans}} = 0.045$) is significantly lower than the Arabic BM25 baseline as well as the Arabic frozen mT5-base. Three structural factors make a direct Arabic–Malay comparison difficult: (1) the datasets differ in task format; (2) XQuAD answer spans are guaranteed passage substrings while Belebele answers may paraphrase; and (3) Malay agglutinative morphology [52], characterized by productive prefixes (*meN-*, *ber-*) and suffixes (*-kan*, *-an*), yields high surface-form diversity per root. Despite these confounding factors, the baseline is the first quantitative retrieval anchor reported publicly for Malay in a RAG-adjacent task to the best of our knowledge.

8.3 Empirical Bound on $\varepsilon_{\text{morph}}$

$$\varepsilon_{\text{morph}} \approx F_{1\text{-ans}}(\text{QLoRA}) - F_{1\text{-ans}}(\text{frozen}) = 0.187 - 0.050 = 0.137 \tag{15}$$

8.4 Stage 2 Interpretation: The O_{context} Paradox

O_{context} measures the proportion of answer tokens that appear verbatim in the retrieved passage (surface-form span overlap). It is a *behavioural diagnostic*, not a hallucination measure: a model can exhibit low overlap while hallucinating (generating tokens absent from the context) or high overlap while remaining fully faithful (copying the exact correct span). This distinction is critical for interpreting Table 7.

Config A (frozen mT5-base, $O_{\text{context}} = 14.5\%$) appears to have the lowest span overlap. This is explained by its generation behaviour: Config A produces long, extractive passages averaging 5.03 words that copy entire phrases verbatim from the context. However, because these phrases are *longer* than the gold span, they fail to match the exact short answer token-for-token, yielding low O_{context} *and* ROUGE-L = 0 simultaneously. The model is doing verbatim extraction of the wrong granularity–copying whole sentences instead of precise answer spans–not superior grounding.

Trained models (Configs B–D) produce compact 1.80–2.12 word answers that target the correct span length. When these answers overlap with the gold span at the token level, O_{context} rises because short precise answers have high overlap density. Config C ($O_{\text{context}} = 60.0\%$) shows 3.5 percentage points less verbatim context overlap than Config D (63.5%). This difference is consistent with the grounding penalty λ_1 reducing verbatim copying pressure–but because O_{context} is a surface-form proxy, this finding is treated as *behavioural evidence* rather than direct validation that Config C

addresses Failure Mode F4. A model can copy less verbatim text while hallucinating more; confirming F4 reduction requires NLI-based semantic faithfulness scoring.

Why O_{context} is a limited metric. O_{context} rewards verbatim extraction and penalizes semantically faithful paraphrase. A model that copies the entire passage scores higher than one that generates a precise, semantically correct answer in different words. For this reason, O_{context} is used here only as a *diagnostic probe* for the pilot, not as a primary evaluation metric. The open-domain follow-up will replace it with CAMeLBERT-NLI-based semantic entailment scoring [25], which measures true faithfulness regardless of surface-form overlap. The pilot findings should therefore be read as: Config C produces *more precisely targeted, less verbatim* answers than Config D, the intended effect of λ_1; direct F4 validation is deferred to the open-domain follow-up.

8.5 Limitations

Six limitations restrict the scope of conclusions and guide the direction of future work:

(1) **Closed-passage scope**: Stage 1 retrieval is evaluated on a 1,190-passage pool where the correct answer is guaranteed to be present. The controlled setup overestimates retrieval performance in the wild, where the relevant passage may not be indexed at all. The actual *R@K* gap for Arabic under open-domain conditions is likely to be larger than the Δ=0.420 gap seen here relative to BM25.

(2) **Stage 2 pilot scale**: Stage 2 uses $n = 400$ test triples and $n = 800$ training examples, which limits statistical power for detecting small effect sizes. The resulting ROUGE-L confidence intervals are relatively wide, and the findings should be interpreted as directional evidence rather than definitive performance estimates.

(3) **O_{context} limitations**: The context-overlap proxy measures surface-form verbatim overlap only; it does not directly measure hallucination or semantic faithfulness. A model can copy less text while hallucinating more, or copy extensively while remaining fully faithful. O_{context} is therefore used as a behavioural diagnostic only. CAMeLBERT-NLI-based semantic entailment scoring is designated for the open-domain follow-up to provide a direct faithfulness signal.

(4) **Unvalidated failure modes**: Failure Modes F2 (context window noise) and F3 (formatting errors) are formally defined and supported by the literature review but lack direct experimental validation in this pilot. Their mathematical metrics (SNR_{gen} and C_{schema}) require multi-passage retrieval and structured output tasks respectively, both of which are reserved for the open-domain follow-up.

(5) **Malay QLoRA absent**: No Malay neural adaptation is reported because Belebele's multiple-choice format does not provide extractive answer spans required for span-extraction fine-tuning. A native extractive Malay QA benchmark analogous to XQuAD is a prerequisite for Stage 1 Malay QLoRA adaptation and represents an identified community resource gap.

(6) **English-only PRISMA search**: The Boolean search queries were formulated in English, which may cause Arabic-medium and Malay-medium venue papers to be missed. This potentially underestimates existing low-resource RAG work published in regional NLP conferences such as those of the Arabic NLP community.

8.6 Future Directions

Building on the empirical evidence presented in this work, a promising direction for future research is the development of **Adaptive Quantized Directional LoRA (AQD-LoRA)**, a hybrid fine-tuning framework designed to address the conservatism effect and open-domain generalization challenges identified in this pilot. Unlike QLoRA, which applies

uniform 4-bit quantization across all transformer layers, AQD-LoRA selectively preserves morphologically sensitive attention layers at full precision (16-bit) while compressing semantically general layers to 4-bit. DoRA-style magnitude–direction decomposition is then applied exclusively to the high-precision layers, enabling fine-grained adaptation of cross-lingual morphological patterns without the memory overhead of full-precision training. This design directly targets the trade-off observed between composite-loss QLoRA (Config C) and full fine-tuning (Config D): AQD-LoRA is hypothesized to recover the length-conservatism reduction achieved by the grounding penalty λ_1 while restoring the expressive capacity suppressed by uniform quantization. Recent work demonstrates that QDoRA–which combines 4-bit quantization with DoRA's magnitude–direction decomposition–matches full fine-tuning accuracy at QLoRA-level memory cost on reasoning tasks [60], providing empirical motivation for AQD-LoRA's selective quantization design in morphologically complex multilingual settings. Empirical validation on open-domain Arabic XQuAD and a native extractive Malay QA benchmark (once such a resource becomes available) represents the natural next experimental step following this pilot.

9 Conclusion

RAG failure modes are now characterized, formalized, and supported by behavioural evidence in a low-resource multilingual setting. This paper has contributed: (1) a PRISMA 2020-conformant systematic review of four canonical RAG failure modes drawn from 61 studies; (2) a two-stage LoRA-based mitigation framework with stage-specific mathematical objectives; and (3) pilot closed-passage QA experiments providing behavioural evidence for the framework on Arabic, alongside what is, to the best of our knowledge, the first cross-lingual quantitative baseline for Malay.

Stage 1 provides preliminary evidence that QLoRA fine-tuning of mT5-base at 0.23% of parameters achieves $F_{1\text{-ans}} = 0.187$ on XQuAD-Arabic, a 272% relative improvement over the frozen baseline at 436× lower parameter cost. Stage 2 shows that composite-loss LoRA (Config C, $\lambda_1 = 0.3$) achieves the highest ROUGE-L in this pilot setting, with Config C showing 3.5 percentage points less verbatim context overlap (O_{context}) than full fine-tuning, treated as behavioural evidence of reduced copying rather than direct F4 validation, which requires NLI-based scoring in the open-domain follow-up. The Malay BM25 baseline motivates the development of a native Malay extractive QA benchmark as a community priority.

Data Availability Statement

XQuAD [2] is available at https://github.com/google-deepmind/xquad (CC-BY 4.0). Belebele [6] is available at https://github.com/facebookresearch/belebele (CC-BY-SA 4.0) and via HuggingFace as `facebook/belebele`, split `zsm_Latn`. mT5-base is available as `google/mt5-base` (Apache 2.0). QLoRA adapter weights and experimental scripts will be deposited upon acceptance.

AI Usage Disclosure

The authors used LLM assistance for grammar checking, sentence-level proofreading, and code formatting. All scientific claims, experimental design, data analysis, mathematical formalizations, and intellectual contributions are solely the authors' own.

Ethics Declaration

This study uses only publicly available benchmark datasets (XQuAD, Belebele) containing no personally identifiable information, under open licences, requiring no institutional ethical approval.

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